



Full length article

A novel seam extraction and path planning method for robotic welding of medium-thickness plate structural parts based on 3D vision

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ABSTRACT

Currently, the robotic welding of medium-thickness plate structural parts has become a common welding application. With the rapid development of automation technology and robotics, the traditional teaching-playback mode and the off-line programming mode cannot meet the automation demand of welding robots. To realize automatic seam extraction and path planning for robotic welding of medium-thickness plate structural parts without programming and teaching, we use three models of medium-thickness plate structural parts as the research objects to propose a novel seam extraction and path planning method for robotic welding of medium-thickness plate structural parts based on 3D vision. Firstly, a set of improved RANSAC multiplanes fitting algorithms is proposed to accurately obtain the position of the intersection lines between the intersecting planes of the point cloud model. On this basis, we combine the geometric features of three models to propose the specific welding seam extraction methods respectively. Then, according to the spatial structure of the welding seams and the welding process, we carry out the welding path planning. Finally, a welding pose planning method based on the dihedral structure is proposed. Experiment results show that the proposed method can well realize the welding seam extraction, welding path and posture planning of medium-thickness plate structural parts without programming and teaching.

1. Introduction

Medium-thickness plates generally refer to steel plates or other metal material plates with thickness greater than 4 mm. Its application fields include construction machinery, steel structure, carriage plate, and container plate. With the rapid development of automation and robot technologies, the robotic welding of medium-thickness plate structural parts has become a common welding application. It usually has the following advantages: stabilize and improve welding quality while increasing labor productivity; improve the labor intensity of workers; reduce the requirements for worker's skills.

When welding medium-thickness plate structural parts with robots, the operator needs to carry out complicated teaching and programming work on the welding path of the welding robot before welding. The “teaching-playback” mode and the offline programming (OLP) are still two main working modes of modern welding robots [1]. Due to the disadvantages of large size, irregular welding seams, and low work-piece clamping accuracy for medium-thickness plate structural parts, the robots are challenging to complete a batch of welding through “teaching-playback” modes. To sum up, the “teaching-playback” and offline programming (OLP) modes are significantly hindering the promotion of robots in welding field.

In recent years, the computer vision aids have gradually been used to solve above problems. The two-dimensional machine vision guidance is a common method, which mainly includes the passive vision method and laser seam tracking method [2–5]. Passive vision uses a camera to capture the weld image and performs image processing on the two-dimensional image to extract the welding path. Xu et al. [6] analyzed the welding spectrum, adjusted the narrow-band filter's central wavelength, and finally obtained a relatively clear picture of the welds with narrow gaps. Gao et al. [7] optimized the dark channel prior anti-interference processing algorithm to remove the metal vapor and plasma sputtering in the welding process and extracted the welding path.

The passive vision method is simple regarding equipment. Nevertheless, it is unsuitable for extracting complex welding seams due to the poor accuracy and lack of three-dimensional information. The laser seam tracking method projects the laser on the plane of the welding seam. Meanwhile, the camera captures the laser pattern from another angle. Finally, the computer extracts the feature points of the cross-section profile in the pattern and connects to obtain the welding

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path [8–15]. Wang et al. [16] proposed a real-time detection and feature extraction algorithm for butt welds based on a laser vision sensor. Zou et al. [17] proposed a welding seam extraction method for butt welds based on a three-line stripe laser vision sensor. **Since the laser structured light is a local-type sensor, it could not perceive the global environment.** Therefore, it is mainly applied to real-time seam tracking and identification but is unsuitable for offline three-dimensional (3D) path planning [18–24]. The aforementioned problems significantly hinder the further application of vision technology in the welding field. They indirectly affect the further popularization of robotic welding for medium-thickness plate structural parts.

With the development of structured light visual sensing technology, researchers gradually use plane structured light sensors to achieve accurate and efficient offline 3D path planning. **Xu et al. [25] proposed a welding path planning method based on the combination of point cloud and deep learning for the problem of highly redundant and uneven dense point clouds to realize automatic welding.** Vrushali Patil et al. [26] completed the robot welding path planning using a plane structured light sensor to take the two space butt plates' point cloud. Yang et al. [27] proposed an algorithm based on the point-to-plane distance to extract V-shaped welding seam feature points. Compared with the laser structured light sensors, the plane structured light sensors have better performance on the 3D reconstruction and global information perception. Therefore, the point cloud processing method for welding robot path planning has become a new solution to welding robots' complex teaching and programming before welding [28–30]. However, the research on welding seam extraction based on 3D vision mainly has the following two problems.

- (1) The research mainly focuses on the automatic extraction of the plane welding seam. There is no research on the automatic extraction of the medium-thickness plate structural parts welding seam.
- (2) Most research is limited to the automatic extraction of a single welding seam. The spatial information of the point cloud is rarely used to extract multiple welding seams simultaneously.

To overcome the above problems, this paper proposes a novel seam extraction and path planning method for robotic welding of medium-thickness plate structural parts based on 3D vision. The main contributions of this paper are as follows.

- (1) The accurate multiplanes fitting of the medium-thickness plate structural parts point cloud model is achieved by the improved RANSAC algorithm. On this basis, we combine the geometric features of the three point cloud models to propose specific welding seam extraction methods respectively.
- (2) We propose a welding path planning method according to the spatial structure of the welding seams and the welding process. Then, a welding posture planning method based on the dihedral structure is proposed to realize the automatic planning of welding torch posture.
- (3) We use real experimental data to evaluate the performance of the proposed method and compare it with the existing method to illustrate its effectiveness.

The paper has the following works. Section 2 introduces the configuration of the experiment system. Section 3 proposes a multiplane fitting method based on the improved RANSAC algorithm. Section 4 offers the specific welding seam extraction method for different medium-thickness plate structural parts. Section 5 proposes the welding path planning method and the welding posture planning method, respectively. Section 6 shows the experimental results and analysis. Section 7 describes the conclusion of this paper.

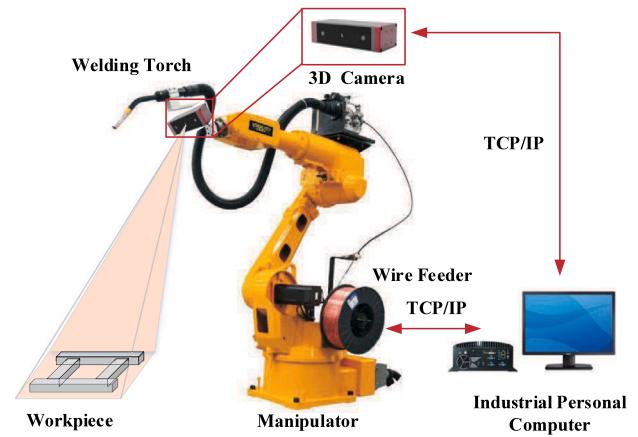


Fig. 1. The robotic welding system.

Table 1

The characteristics of 3D camera.

Case	Parameters	Value
1	Working Principle	Binocular Structured Light
2	Optimum Working Distance (mm)	500 ± 250 mm
3	Maximum Working Distance	750
4	Field of View(FOV)	H52° × V31°
5	Repeat Accuracy	±0.5 mm

2. The experiment system and framework

2.1. Experiment system

As shown in Fig. 1, the experimental system divides into the welding execution and 3D vision systems. The welding execution system is composed of a welding torch, a wire feeder, a manipulator, and a robot controller. The combination of a 3D camera and manipulator adopts the **Eye-In-Hand** method.

The 3D vision system includes a 3D surface scanning structured light camera and an industrial personal computer (IPC). The 3D camera scans the workpiece to form a point cloud and transmits it to the IPC through the TCP/IP protocol. The IPC performs algorithm processing on the obtained point cloud to extract the robotic welding path. Finally, the IPC converts the welding path into the scripting language of the robot and transmits it to the robot controller through the TCP/IP protocol.

Fig. 2 shows the 3D surface scanning industrial camera used in this paper, which adopts grating projection technology to measure objects. The structured light projector of the camera projects a sinusoidal laser fringe onto the measured object. Then the CCD camera captures the measured object and obtains the phase by demodulating the captured laser fringe. Finally, the camera calculates the phase to get the depth information. Table 1 shows the parameters of the 3D camera.

2.2. Experiment model

This paper uses three kinds of medium-thickness plate structural parts as research models. As shown in Fig. 3, these models mainly include the three-plane butt, the H-section steel butt, and the square tube butt, which are commonly used in the industrial production of compartment panels, steel structures, civil air defense doors, and containers [31–33]. The three models need to be welded twice in the industrial production process. In the first time, the rough welding is performed to fix the position of the medium-thickness plate structural parts. This process only requires the welding worker to use spot welding to assemble the medium-thickness plate structural parts, which is not the focus of this paper. The second time is the advanced welding

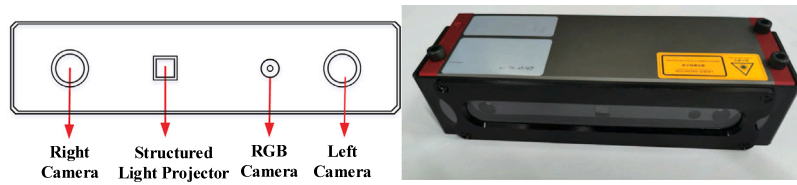


Fig. 2. The 3D surface scanning industrial camera.



Fig. 3. Three medium-thickness plate structural parts models.

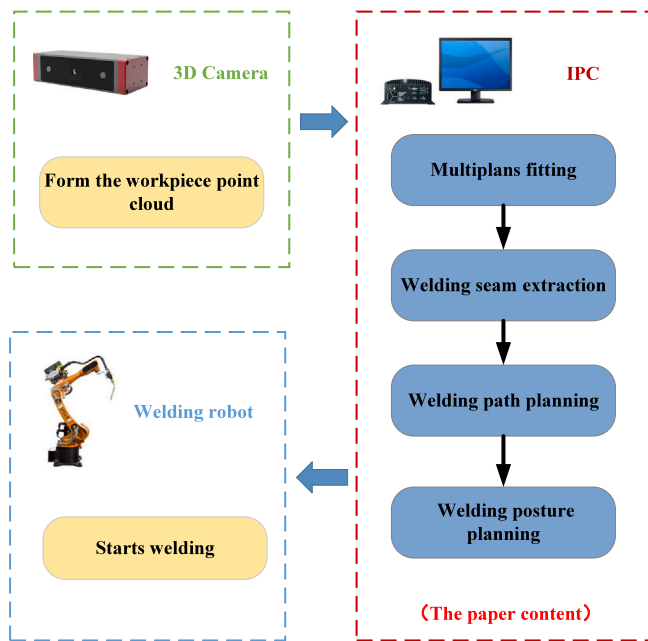


Fig. 4. The system framework.

to assemble the medium-thickness plate structural parts [34]. In this process, the quality of welding will determine the quality of the final industrial product. Therefore, the robotic and automatic welding of this process without programming and teaching is the core problem to be solved in this paper.

2.3. System framework

The framework of the proposed method is shown in Fig. 4.

1. **Multiplans fitting:** A set of improved RANSAC multiplanes fitting algorithms is proposed to ensure the accuracy of welding seam extraction.
2. **Welding seam extraction:** According to the geometrical characteristics of medium-thickness structural parts, a specific method is proposed for the welding seam extraction of three different medium-thickness structural parts, which can extract multiple welding seams simultaneously and ensure the accuracy of welding seam extraction.

3. **Welding path planning:** To ensure the welding quality, determine the arc starting point and arc extinguishing point of each welding seam according to the welding process.
4. **Welding posture planning:** The welding posture planning is also a critical factor in affecting the welding quality. A welding posture planning method based on the dihedral structure is proposed to solve the pose of the welding torch.

3. Multiplans fitting

Medium-thickness plate structural parts are usually composed of standard planes. So it is necessary to carry out multiplane fitting of the workpiece point cloud before welding by the proposed method to accurately extract the welding seam and plan the welding path in subsequent. At present, there are many types of multiplane fitting algorithms, but their core mainly relies on two categories: the clustering algorithm and the RANSAC algorithm. The clustering algorithm is suitable for the surface recognition of rough fields, which can identify and separate a large number of planes in a short time, but the process has low precision, so it is not suitable for the precision requirements of the welding process. At the same time, in industrial welding scenarios, to be suitable for small batches and multi-variety workpieces, an algorithm with few adjustable parameters should be used to improve the method's robustness. The multiplanes fitting method based on clustering requires parameter adjustment according to the workpiece type, while the RANSAC algorithm does not. Therefore, on the premise of ensuring welding efficiency, stability and accuracy, we choose the RANSAC algorithm as the basic algorithm for multiplane fitting and improve it according to the actual needs of welding.

Fig. 5 shows the principle of the RANSAC algorithm. The original data sets are represented by the blue points in Fig. 5(a). The size of the minimum number set for the estimated plane model is 3, so we select three points in the original data set to calculate the plane model. Then, as shown in Fig. 5(b), we use the model to match the original data set and calculate the distance from each data point to the plane model. We set the distance threshold according to the requirements of the plane fitting accuracy. Points with a distance less than the threshold are regarded as inliers, and points with a distance greater than the threshold are outliers. As shown in Fig. 5(c), the RANSAC algorithm iteratively processes the data and takes the plane model with the most inliers as the final fitting plane.

3.1. Experimental analysis of RANSAC algorithm

The RANSAC plane fitting algorithm stops iterating when it satisfies one of the following three conditions: reaching a fixed number of

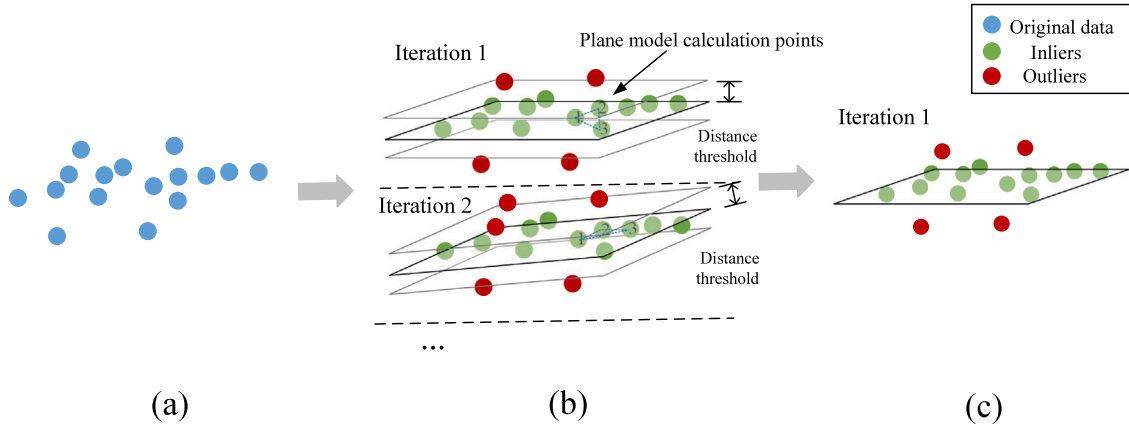


Fig. 5. The principle of the RANSAC algorithm.

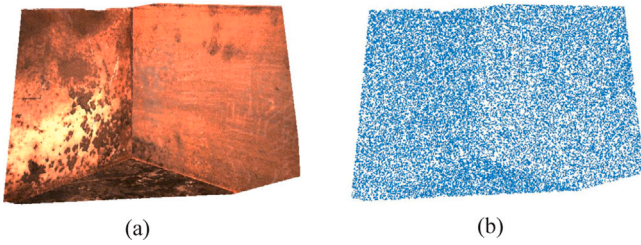


Fig. 6. The three-plane butt model point cloud.

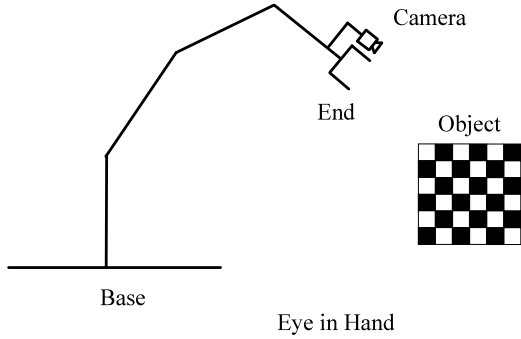


Fig. 7. The principle of Eye-In-Hand calibration.

iterations, or a fixed number of interior points, or both the number of iterations and interior points. The RANSAC method has high stability in fitting a single plane with appropriate parameter settings. Nevertheless, it has defects in multiplanes fitting. We analyzed its defects by the multiplanes fitting of the three-plane butt point cloud.

Fig. 6(a) shows the three-plane butt point cloud model. Due to a large amount of data, the model is down-sampled. Fig. 6(b) shows the model after down-sampling with the points reduced from 971877 to 59156.

We accomplished a multi-plane fitting experiment using the above three conditions for the down-sampled model. According to the imaging point accuracy of the 3D camera, the distance threshold is set to 1 mm by trial and error.

In the Eye-in-Hand calibration method shown in Fig. 7, for any two poses in the robot movement process, the following Eq. (1) holds

$$\begin{aligned} & \text{Base}T_{\text{End}2} \times \text{End}2T_{\text{Camera}2} \times \text{Camera}2T_{\text{Object}} \\ &= \text{Base}T_{\text{End}1} \times \text{End}1T_{\text{Camera}1} \times \text{Camera}1T_{\text{Object}} \end{aligned} \quad (1)$$

The external matrix $\text{End}T_{\text{Camera}}$ with the smallest error is selected according to Eq. (1) after multiple calibrations. Then the point cloud

model coordinates in the robot coordinate system are solved when the camera works according to the 3D camera shooting pose and external matrix.

For condition one, we set the number of iterations from 100 to 600 in increments of 50. Figs. 8(a) and 8(b) show the experimental results of 300 iterations with a high error rate and 600 iterations with better stability after many experiments.

For condition two, we set the number of interior points from 5000 to 10000 in increments of 1000. Figs. 8(c) and 8(d) show the experimental results of 5000 interior points and 10000 interior points after many experiments.

For condition three, it satisfies the above two conditions simultaneously. As shown in Figs. 8(e) and 8(f), the desired effect was unachieved after many experiments.

3.2. The improved RANSAC algorithm

Through the experimental analysis, there are two main problems in the RANSAC multiplanes fitting algorithm.

- (1) The algorithm needs too many iterations, resulting in low efficiency.
- (2) The setting of the distance threshold by the trial and error method will directly affect the accuracy of the multiplanes fitting.

For the first problem, as shown in Fig. 9, the algorithm can identify the plane better when the three model calculation points are calculated in the same plane.

The medium-thickness plate structural parts consist of n planes, and each plane contains the same number of data points. Let $\text{Cloud} = \{C_i, i = 1, 2, \dots, n\}$ be the 3D point cloud model, the algorithm randomly selects one point $C_1(x_1, y_1, z_1)$ as the calculation point. The probability that the remaining two calculation points $C_2(x_2, y_2, z_2)$ and $C_3(x_3, y_3, z_3)$ are selected on the same face is:

$$P_{\text{one_plane}} = 1 \times \frac{1}{n} \times \frac{1}{n} = \frac{1}{n^2} \quad (2)$$

According to Eq. (2), as the number of planes increases, the number of iterations becomes n^2 times to ensure that the algorithm selects three model calculation points in the same plane. As a result, the random selection of model calculation points causes too many iterations.

For the second problem, the setting of the distance threshold affects the number of interior points in each plane. When the algorithm completes the fitting of a plane and segments its interior points, the remaining point cloud data will directly affect the fitting accuracy of the remaining plane. Therefore, a complete and accurate segmentation of each fitted plane point cloud becomes the core of improving the accuracy of multiplanes fitting.

To overcome the problems of the above RANSAC algorithm for multiplanes fitting, we improve it in two aspects.

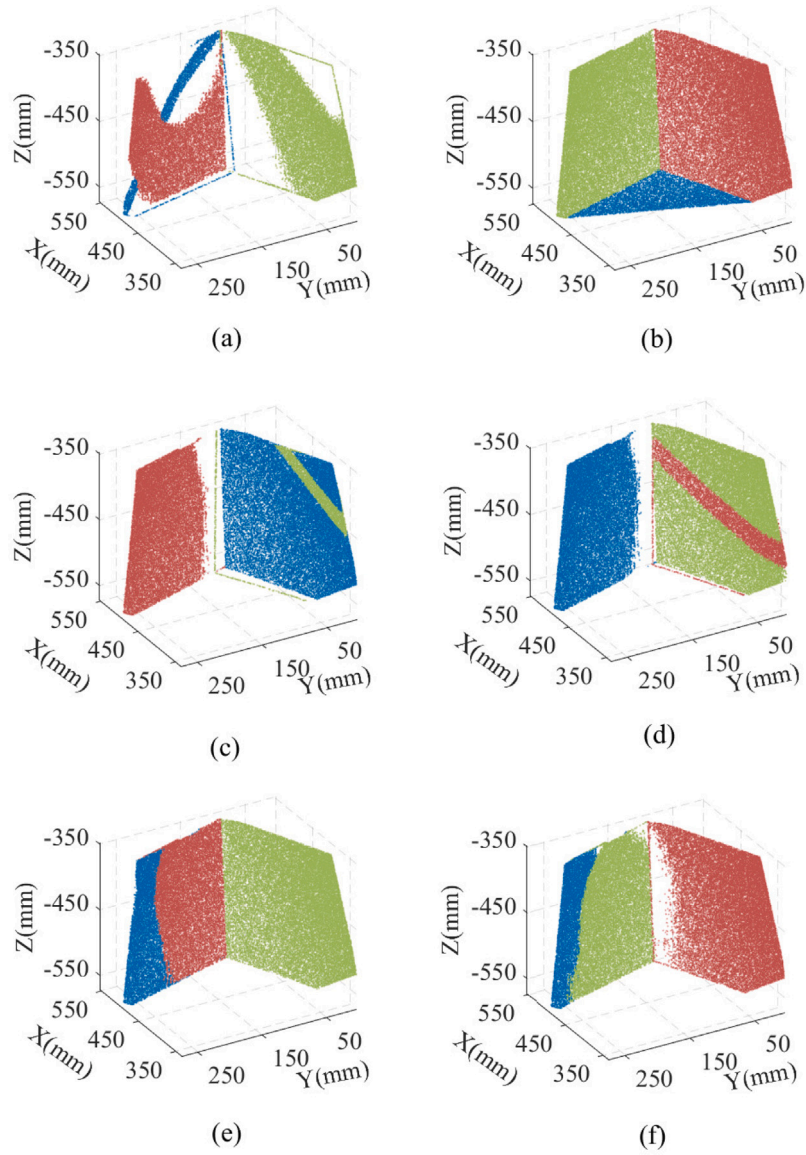


Fig. 8. The experimental results.

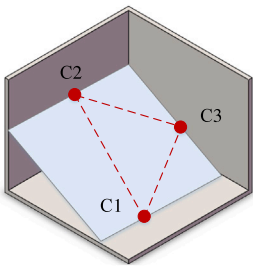


Fig. 9. Selection the model calculation points.

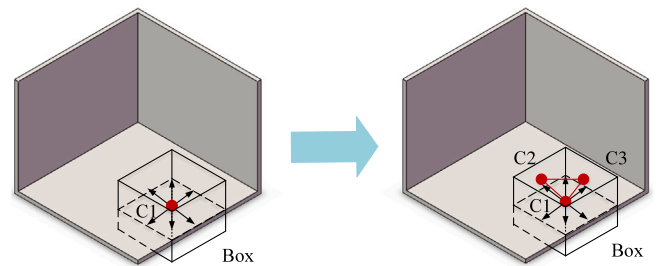
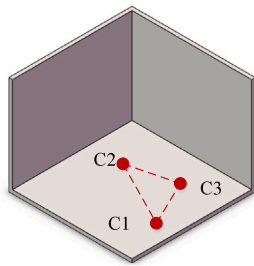


Fig. 10. The bounding box.

- (1) A model calculation point selection method based on K-nearest neighbors is proposed to reduce the number of algorithm iterations for multifaceted fitting.
- (2) An accurate quadratic plane fitting algorithm is proposed to improve the fitting accuracy of a single plane.

3.2.1. Selection method of the model calculation points based on K-nearest neighbors

As shown in Fig. 10, we choose one of the points as the first calculation point of the plane model. The original method needs to traverse all the points and find the nearest neighbor point by solving

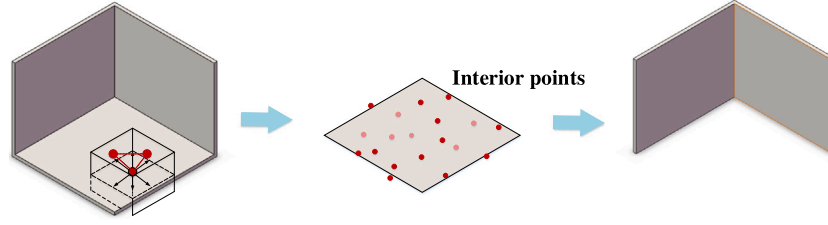


Fig. 11. The rough fitting and segmentation of multiplanes.

Table 2

The experimental results of distance threshold selection.

Case	Distance threshold (mm)	Number of failures
1	5	3
2	10	0
3	20	7
4	40	13

Table 3

The experimental results of the selection of the number of iterations.

Case	Iterations	Running time (ms)	Number of failures
1	5	648	2
2	10	1203	0
3	15	2012	0

the distance from all points to the center point according to Eq. (3).

$$D = \sum_{i=1}^n \sqrt{(x_{c_i} - x_1)^2 + (y_{c_i} - y_1)^2 + (z_{c_i} - z_1)^2} \quad (3)$$

To speed up the calculation process, according to the density of the point cloud, we take the first calculated point as the center point to establish the bounding box. Then, we find the nearest neighbor points on the points inside the bounding box.

To select a reasonable bounding box distance threshold, we make the three-plane butt model as the experimental object, and set the bounding box distance threshold as 5 mm, 10 mm, 20 mm and 40 mm. We conducted 30 times computational experiments for each group, and the experimental results are shown in Table 2.

After experiments, we conclude that when the box distance threshold is set too large, it is easy to include the point clouds of other planes, resulting in the failure of the calculation and the decrease of the robustness of the algorithm. When the box distance threshold is set too small, there are fewer points in the box, and due to the fluctuating distribution of the point cloud, a small number of points cannot represent the features of the plane. Taking all into consideration, we take 10 mm as the distance threshold and then establish the bounding box as

$$Box = \begin{cases} x_{Box} \in \{x_1 \pm 10\} \\ y_{Box} \in \{y_1 \pm 10\} \\ z_{Box} \in \{z_1 \pm 10\} \end{cases} \quad (4)$$

After statistics, the number of points in the bounding box is usually 60–80. To stably find the nearest neighbor points of the target number, we set the search number of nearest neighbor points to 50. Finally, the second and third model calculation points are randomly selected among the nearest neighbor points.

Fig. 11 shows the rough plane fitting of the model after selecting the model calculation points. After rough fitting, the plane points are segmented by setting a larger distance threshold. Finally, the improved algorithm fits all planes by repeating the above steps.

This method can arbitrarily select a point as one of the model calculation points and calculate the rough fitting equation of the plane. Therefore, it can solve the problem of too many iterations in the original RANSAC algorithm. After many experiments, a particular case is that the algorithm selects the first model calculation point at the intersection of the planes, resulting in inaccurate plane recognition. To avoid this problem, we need to increase the number of iterations of the algorithm appropriately. At the same time, we need to satisfy the requirements of algorithm efficiency. Therefore, we conducted experiments on the selection of the number of iterations. We make the three-plane butt model as the experimental object, and set the number

of iterations to 5, 10, and 15, and conducted 30 times computational experiments for each group. The experimental results are shown in Table 3.

The experimental results show that when the number of iterations is 10 times, it can satisfy the efficiency of the algorithm and avoid the occurrence of the particular case. Therefore, the number of iterations is set to 10 times.

Affected by the accuracy of the imaging point of the camera and model plane roughness, the fluctuation range of the point cloud in the same plane is about ± 1 mm. Since the current is a rough plane fitting process, to ensure the accurate fitting in the next step, the algorithm should keep the features of the current model plane point cloud as much as possible. That is, the points on the model plane point cloud should be identified as inliers as much as possible. Therefore, the setting of the distance threshold should be greater than 1 mm.

After many tests, we found some points are not recognized as inliers when the distance threshold is set to 1.5 mm. When it is set to 2 mm, all points can meet the above requirements except for the points with a larger deviation from the plane point cloud. Finally, we set 2 mm as the distance threshold.

3.2.2. An accurate quadratic plane fitting algorithm based on centroid

After the rough fitting and segmentation of the multiplanes, we have obtained the independent point cloud data of each plane. Due to the rough segmentation, each plane will be over-segmented. As shown in Fig. 12, the black dividing line depicts an accurate three-plane dividing line. However, rough segmentation is a way of over-segmentation, which will regard points in other planes that meet the distance threshold as interior points. In this case, the segmented plane points need to perform an accurate quadratic fitting on the roughly segmented plane.

Fig. 13 is the top view of the red plane in Fig. 12. The points in the blue circle in Fig. 13 will directly affect the accurate fitting of the plane.

To accurately select the model calculation point, we choose the centroid C_c of the plane, defined in Eq. (5), as the first model calculation point, where denotes the number of points.

$$C_c = (x_c, y_c, z_c) = \frac{1}{n} \left(\sum_{i=0}^n x_i, \sum_{i=0}^n y_i, \sum_{i=0}^n z_i \right) \quad (5)$$

The remaining two model calculation points are selected through the K-nearest neighbors method. The normal vector of the plane is

$$\vec{n} = (x_n, y_n, z_n) = \vec{C_2 C_c} \times \vec{C_3 C_c} \quad (6)$$

According to Eq. (7), the equation of the plane in the robotic coordinate system is

$$F(x, y, z) = x_n(x - x_{C_c}) + y_n(y - y_{C_c}) + z_n(z - z_{C_c}) \quad (7)$$

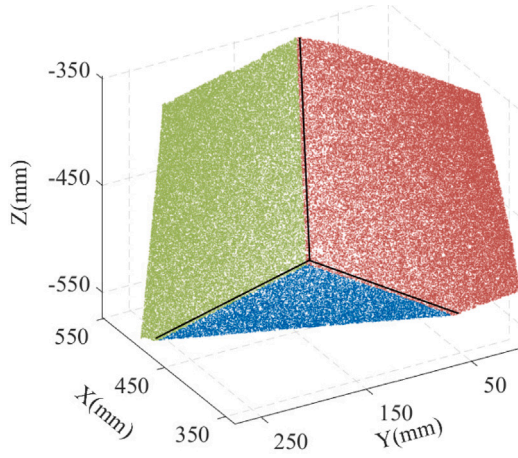


Fig. 12. The rough plane fitting results.

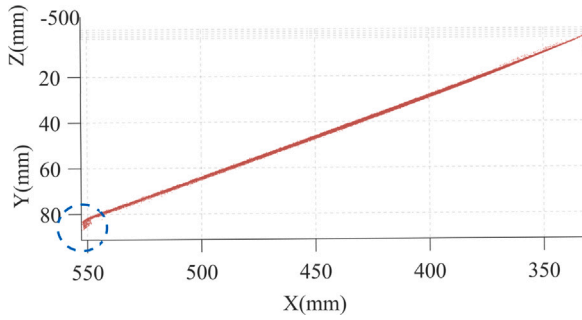


Fig. 13. The top view of the red plane in Fig. 11.

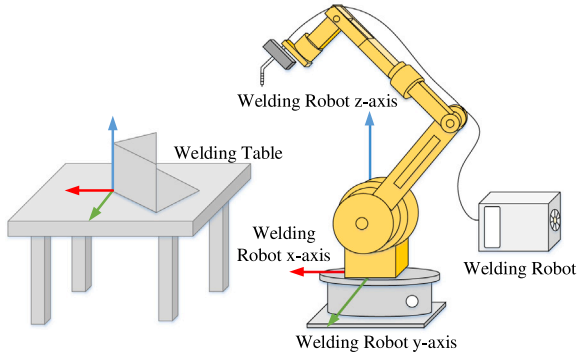


Fig. 14. The position of the three-plane butt model in the robot coordinate system.

3.3. Rearrange the fitted plane

Due to the randomness of the fitting surfaces of the improved RANSAC algorithm, the algorithm cannot guarantee the fitting order of each plane for each run. However, there are strict requirements on the sequence of welding seam welding in the welding process. Therefore, we need to rearrange each fitted plane according to the set order. This paper takes the three-plane connection model as an example. **Due to the limitation of the welding process in the industrial field, the three-plane butt model is chiefly horizontally placed.**

As shown in Fig. 14, we can reorder the three fitted planes according to the distribution of point cloud coordinates. Since the bottom plane of the three-plane butt model has distributed on the same horizontal plane along the z-axis of the robot coordinate system, the plane with the lowest z-coordinate value of the centroid is taken as the bottom plane of the three-plane docking model and marked as the first plane.

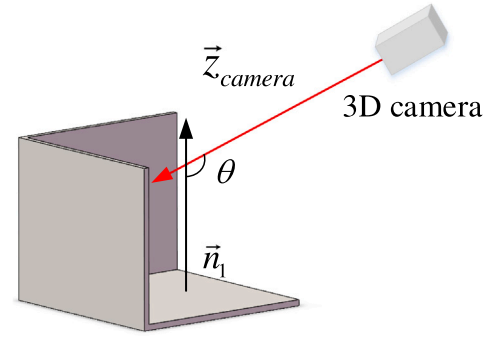


Fig. 15. The shooting posture of the 3D camera.

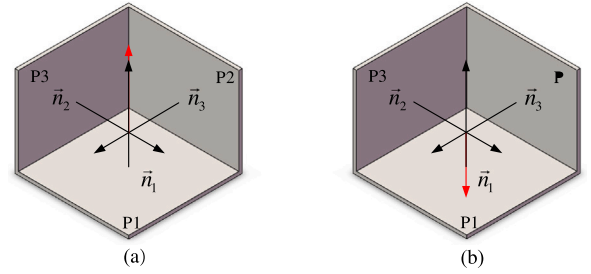


Fig. 16. The second and third planes definition diagram.

We sort the remaining two faces in the camera coordinate system. The normal vector $\vec{n}_1$ of the first plane in the robot coordinate system is obtained by Eq. (7).

Fig. 15 shows the shooting pose of the 3D camera. At this time, there is a certain angle between the camera z-axis $\vec{z}_{camera}(0, 0, 1)$ and $\vec{n}_1$. The angle obtained by

$$\begin{cases} \langle \vec{z}_{camera}, \vec{n}_1 \rangle = \frac{\vec{z}_{camera} \cdot \vec{n}_1}{|\vec{z}_{camera}| |\vec{n}_1|} \\ \theta = \arccos \langle \vec{z}_{camera}, \vec{n}_1 \rangle \end{cases} \quad (8)$$

The normal vector $\vec{n}_1$ of the first plane is redefined by

$$\vec{n}_1 = \begin{cases} (x_{n_1}, y_{n_1}, z_{n_1}) & \text{if } \theta > 0 \\ (-x_{n_1}, -y_{n_1}, -z_{n_1}) & \text{if } \theta \leq 0 \end{cases} \quad (9)$$

Based on the similar ways, we can acquire the normal vectors of the remaining two planes, denoted by $\vec{n}_2(x_{n_2}, y_{n_2}, z_{n_2})$ and $\vec{n}_3(x_{n_3}, y_{n_3}, z_{n_3})$, respectively. The outer product of $\vec{n}_2$ and $\vec{n}_3$ is

$$\vec{m} = \vec{n}_2 \times \vec{n}_3 \quad (10)$$

By comparing the directions of $\vec{m}$ and $\vec{n}_1$, we define

$$\begin{cases} \frac{x_{n_1}}{y_{n_2} z_{n_3} - z_{n_2} y_{n_3}} > 0, \vec{m} \text{ and } \vec{n}_1 \text{ are in the same direction} \\ \frac{x_{n_1}}{y_{n_2} z_{n_3} - z_{n_2} y_{n_3}} \leq 0, \vec{m} \text{ and } \vec{n}_1 \text{ are in the opposite direction} \end{cases} \quad (11)$$

As shown in Fig. 16(a), when $\vec{m}$ and $\vec{n}_1$ are in the same direction, the plane equation corresponding to $\vec{n}_2$ describes the second plane, and the plane equation corresponding to $\vec{n}_3$ is the third plane. Fig. 16(b) shows the otherwise result.

The first to third planes of the other two models can be found in the same way. For the H-section steel butt model, the last remaining face is the fourth plane. For the square tube butt model, we obtain the centroids of the remaining two planes. The smaller z-coordinate value of the centroids is the fourth plane, and the other is the fifth plane.

Fig. 17 shows the final result of the multiplanes fitting of the models through the improved RANSAC algorithm.

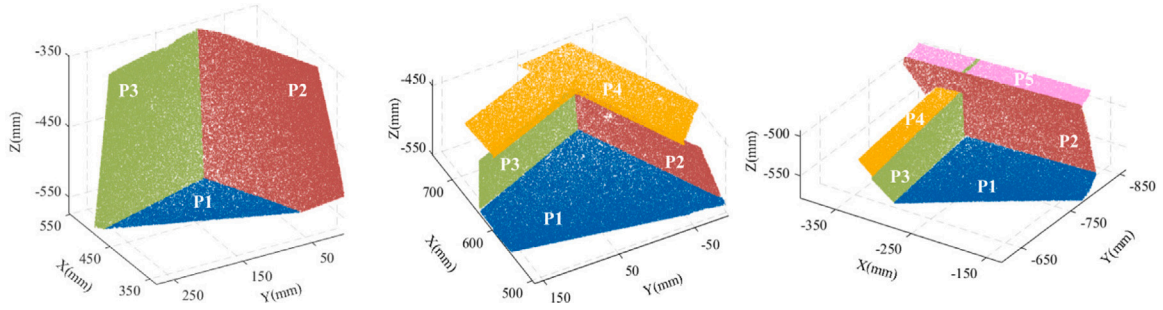


Fig. 17. The multiplanes fitting results based on the improved RANSAC algorithm.

4. Welding seam extraction

The multiplanes fitting of three medium-thickness plate structural models by the improves RANSAC algorithm. On this basis, we adopt the idea of finding the **intersection line between the intersecting planes**, analyzing the geometric structure of each structural part, and finally realize the welding seam extraction.

4.1. The welding seam extraction of the three-plane butt model

After reordering the fitted planes, the point cloud at the intersection line is extracted based on the principle of solving the intersection line of planes. Then, the intersection line point cloud is fitted with the straight-line fitting method based on singular value decomposition (SVD). The straight-line must pass through the location of the arithmetic average of the coordinates $(\bar{x}, \bar{y}, \bar{z})$. Therefore, we take **the arithmetic mean of the point cloud for each intersecting line**. The difference matrix A between the coordinates of each point and the arithmetic average is calculated by

$$A = \begin{bmatrix} x_1 - \bar{x} & y_1 - \bar{y} & z_1 - \bar{z} \\ x_2 - \bar{x} & y_2 - \bar{y} & z_2 - \bar{z} \\ x_3 - \bar{x} & y_3 - \bar{y} & z_3 - \bar{z} \\ \vdots & \vdots & \vdots \\ x_n - \bar{x} & y_n - \bar{y} & z_n - \bar{z} \end{bmatrix} \quad (12)$$

The singular value decomposition of matrix A is performed by

$$A^T A = (V S^T U^T) U S V^T = V S^T (U^T U) S V^T \\ = V (S^T S) V^T = V \begin{bmatrix} \sigma_{\max}^2 & & \\ & \ddots & \\ & & \sigma_{\min}^2 \end{bmatrix} V^T \quad (13)$$

U is an $n \times n$ orthogonal matrix. S is a square matrix consisting of r singular values arranged diagonally from large to small along the arranged diagonally, and r is the rank of matrix A. V is a 3×3 singular vector matrix arranged from large to small along the column direction. The direction of the obtained line is the same as the singular vector corresponding to the maximum singular value. Therefore, the first column of the V matrix is selected as the direction of the fitted straight line, and the singular vector matrix with three rows and one column is expressed as V_d .

We can define a straight line by the known direction and one point. The coordinates (x_l, y_l, z_l) of all points on this straight line satisfy Eq. (14). t is the relation variable between the point coordinates on the straight line and the arithmetic average of all sample point coordinates.

$$\begin{cases} x_l = \bar{x} + V_d(1, 1) \times t \\ y_l = \bar{y} + V_d(1, 2) \times t \\ z_l = \bar{z} + V_d(1, 3) \times t \end{cases} \quad (14)$$

Fig. 18 shows the fitting of three-plane butt model intersection line points based on SVD.

According to the welding sequence, the intersection of the second and third planes is considered as the first welding seam. The first and

second planes formulates the second welding seam. The third welding seam is defined by the first and third planes. Since each welding seam has an endpoint with roughly the same coordinate value, we take the average of these three endpoints and define it as the intersection of the three welding seams. Finally, Fig. 19 shows the welding seams extraction result of the three-plane butt model.

4.2. The welding seam extraction of the H-section steel butt model

As shown in Fig. 20, we find the intersection line of the second and third planes by the straight-line fitting method based on SVD and obtain the intersection point A of this line with the fourth plane and the intersection point B of the first plane.

This paper uses the standard 10-gauge H-beam with a width of 68 mm. Similarly, according to the different specifications of the H-beam, we assume its width is b mm. Therefore, the point D is obtained by shifting $b/2$ mm from the point A in the $\bar{n}_2$ direction. Similarly, the point F is obtained by shifting $b/2$ mm from the point B in the $\bar{n}_2$ direction. The coordinates of points D, F are

$$\begin{cases} (x_D, y_D, z_D) = \\ (b/2 \cdot x_{\bar{n}_2} + x_A, b/2 \cdot y_{\bar{n}_2} + y_A, b/2 \cdot z_{\bar{n}_2} + z_A) \\ (x_F, y_F, z_F) = \\ (b/2 \cdot x_{\bar{n}_2} + x_B, b/2 \cdot y_{\bar{n}_2} + y_B, b/2 \cdot z_{\bar{n}_2} + z_B) \end{cases} \quad (15)$$

Similarly, the coordinates of points C, E and G are

$$\begin{cases} (x_C, y_C, z_C) = \\ (b/2 \cdot x_{\bar{n}_3} + x_D, b/2 \cdot y_{\bar{n}_3} + y_D, b/2 \cdot z_{\bar{n}_3} + z_D) \\ (x_E, y_E, z_E) = \\ (-b/2 \cdot x_{\bar{n}_3} + x_D, -b/2 \cdot y_{\bar{n}_3} + y_D, -b/2 \cdot z_{\bar{n}_3} + z_D) \\ (x_G, y_G, z_G) = \\ (b/2 \cdot x_{\bar{n}_3} + x_F, b/2 \cdot y_{\bar{n}_3} + y_F, b/2 \cdot z_{\bar{n}_3} + z_F) \end{cases} \quad (16)$$

The points M and N is obtained by adjusting the length of AB. The coordinates of points M and N is

$$\begin{cases} (x_M, y_M, z_M) = \\ \left(\frac{1}{6} |\overline{AB}| \cdot x_{\overline{AB}} + x_A, \frac{1}{6} |\overline{AB}| \cdot y_{\overline{AB}} + y_A, \frac{1}{6} |\overline{AB}| \cdot z_{\overline{AB}} + z_A \right) \\ (x_N, y_N, z_N) = \\ \left(\frac{1}{6} |\overline{AB}| \cdot x_{\overline{BA}} + x_B, \frac{1}{6} |\overline{AB}| \cdot y_{\overline{BA}} + y_B, \frac{1}{6} |\overline{AB}| \cdot z_{\overline{BA}} + z_B \right) \end{cases} \quad (17)$$

After the above calculations, Fig. 21 shows the welding seams extraction result of the H-section steel butt model.

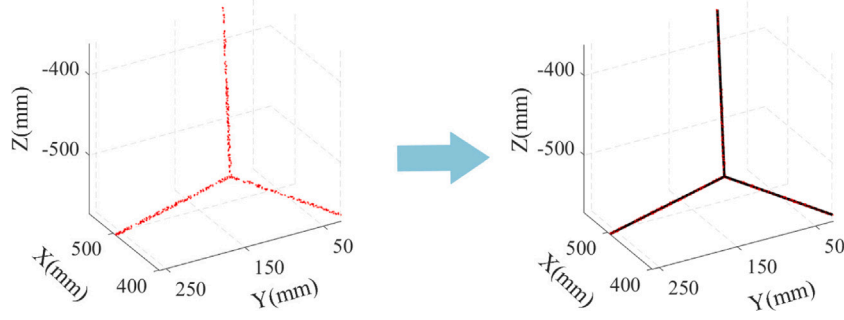


Fig. 18. The fitting of intersection line points based on SVD.

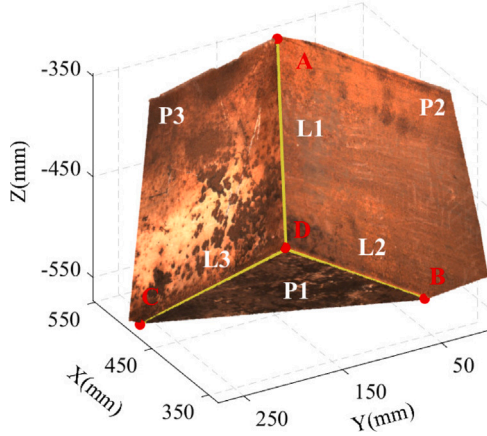


Fig. 19. The welding seams extraction result of the three-plane butt model.

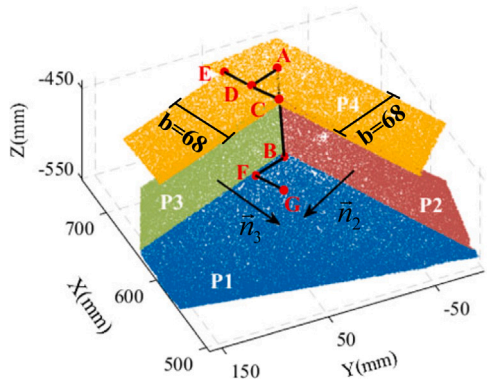


Fig. 20. The welding seams extraction of the H-section steel butt model.

4.3. The welding seam extraction of the square tube butt model

Fig. 22 shows all intersection lines obtained by the straight-line fitting method based on SVD except the third and fourth planes and represents the endpoints of the intersection lines by the letters. The coordinates of the intersection point A of the line segment BC and DE is solved by

$$\begin{cases} (point_{BC}, point_{DE}) = \\ \min(\sum_{i \in BC} \sum_{j \in DE} \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2}) \\ (x_A, y_A, z_A) = \\ (\frac{x_{point_{BC}} + x_{point_{DE}}}{2}, \frac{y_{point_{BC}} + y_{point_{DE}}}{2}, \frac{z_{point_{BC}} + z_{point_{DE}}}{2}) \end{cases} \quad (18)$$

Table 4

Welding path planning for three-plane butt model.

Welding sequence	Welding arc starting point	Welding arc extinguishing point
L1	A	D
L2	D	B
L3	D	C

Table 5

Welding path planning for H-section steel butt model.

Welding sequence	Welding arc starting point	Welding arc extinguishing point
L1	M	N
L2	E	C
L3	B	F
L4	F	G

Table 6

Welding path planning for square tube butt model.

Welding sequence	Welding arc starting point	Welding arc extinguishing point
L1	A	B
L2	A	G
L3	B	F
L4	B	H

In the same way, the coordinates of the intersection point B of the three lines can be solved.

This paper uses the standard square tube with a width of 30 mm. Similarly, according to the different specifications of square tube, we assume its width is d mm. Therefore, the point H is obtained by shifting d mm from the point A in the $\vec{n}_3$ opposite direction. The coordinates of H is

$$\begin{aligned} (x_H, y_H, z_H) = \\ (-d \cdot x_{\vec{n}_3} + x_A, -d \cdot y_{\vec{n}_3} + y_A, -d \cdot z_{\vec{n}_3} + z_A) \end{aligned} \quad (19)$$

After the above calculations, Fig. 23 shows the welding seams extraction result of the square tube butt model.

5. Welding path and posture planning

5.1. Welding path planning

According to Figs. 19, 21, and 23, we carry out welding path planning for the three paper models and show the results in Tables 4, 5, and 6.

According to the actual structure of the model, we can selectively extract the welding seam of the model. When the welding seam length needs to be adjusted, the L1 in the three-plane butt model as an example to illustrate.

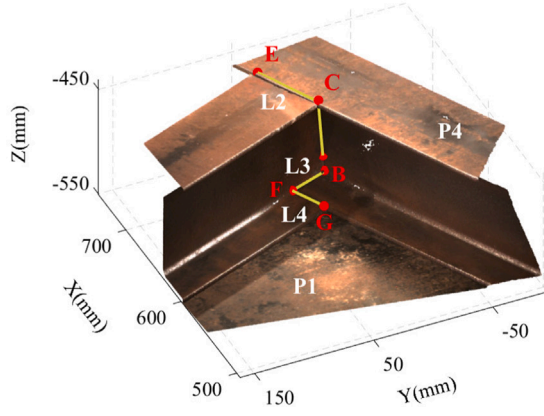


Fig. 21. The welding seams extraction result of the H-section steel butt model.

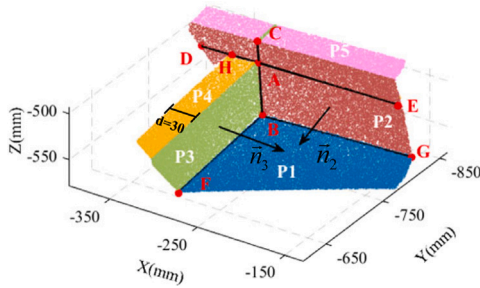
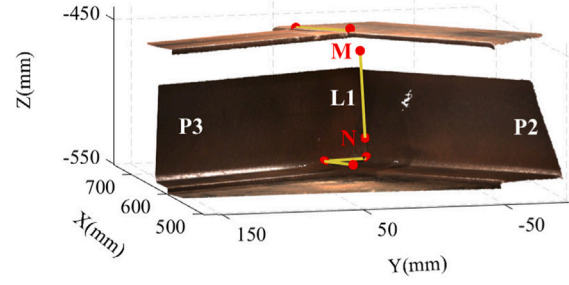


Fig. 22. The welding seams extraction of the square tube butt model.

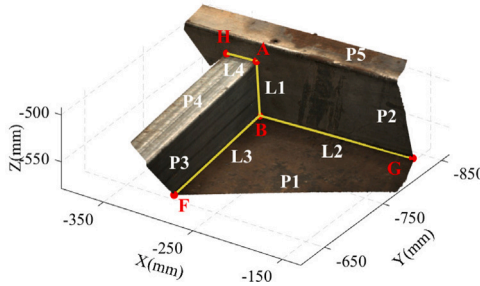


Fig. 23. The welding seams extraction result of the square tube butt model.

The welding direction vector is set according to the arc starting point and the arc extinguishing point:

$$\overrightarrow{AD} = (x_D - x_A, y_D - y_A, z_D - z_A) \quad (20)$$

The welding direction vector unitization is

$$\overrightarrow{AD}' = \frac{\overrightarrow{AD}}{|\overrightarrow{AD}|} \quad (21)$$

If the welding length is L mm, the coordinates of the arc extinguishing point is

$$(x, y, z) = \left(L \cdot x_{\overrightarrow{AD}'} + x_A, L \cdot y_{\overrightarrow{AD}'} + y_A, L \cdot z_{\overrightarrow{AD}'} + z_A \right) \quad (22)$$

The L1 takes point A as the welding arc starting point and the coordinates obtained by Eq. (22) as the arc extinguishing point. The actual welding length adjustment principle of other welding seams is the same as L1.

5.2. Welding posture planning

The welding seam position, axis, and centerline direction are the essential characteristics of the welding seam. The welding torch posture is relative to the welding seam. When the end of the welding torch coincides with the welding point, and the axis coincides with the welding seam centerline, Fig. 24 describes the welding torch posture and the welding seam characteristics by the dihedral structure.

The welding seam coordinate system is $X_s Y_s Z_s$. The X_s coincides with the welding seam axis, and the Z_s coincides with the welding seam centerline. The Y_s solved by the right-hand rule. The X_s solved by

$$X_s = \frac{\frac{\partial x}{\partial t} i + \frac{\partial y}{\partial t} j + \frac{\partial z}{\partial t} k}{\left\| \frac{\partial x}{\partial t} i + \frac{\partial y}{\partial t} j + \frac{\partial z}{\partial t} k \right\|} \quad (23)$$

As shown in Fig. 25, we take the welding seam as the centerline and select the point cloud in the surrounding W mm range. Then, we select the edge point cloud according to the principle of the farthest point from the welding seam within the selected point cloud range and fit it with a straight line. Finally, we solve the parallel edges lines $S1$ and $S2$ of the welding seam.

We perform equidistant interpolation on welding seam, $S1$, and $S2$, then connect the interpolation points of $S1$ and $S2$ at each interpolation point P to get $\vec{n}_r$ and $\vec{n}_l$. Finally, the Z_s is defined by

$$Z_s = \frac{\vec{n}_r + \vec{n}_l}{\|\vec{n}_r + \vec{n}_l\|} \quad (24)$$

The welding torch coordinate system $X_g Y_g Z_g$ follows the convention of

$$\begin{cases} X_g = X_s \\ Z_g = -Z_s \\ Y_g = X_g \times Z_g \end{cases} \quad (25)$$

According to the coordinates of each interpolation point P and the $X_g Y_g Z_g$, the homogeneous matrix can be obtained by

$$T_g = \begin{bmatrix} x_{X_g} & x_{Y_g} & x_{Z_g} & x_p \\ y_{X_g} & y_{Y_g} & y_{Z_g} & y_p \\ z_{X_g} & z_{Y_g} & z_{Z_g} & z_p \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (26)$$

At last, the Euler angle of the welding posture is solved by the homogeneous matrix at the current welding point.

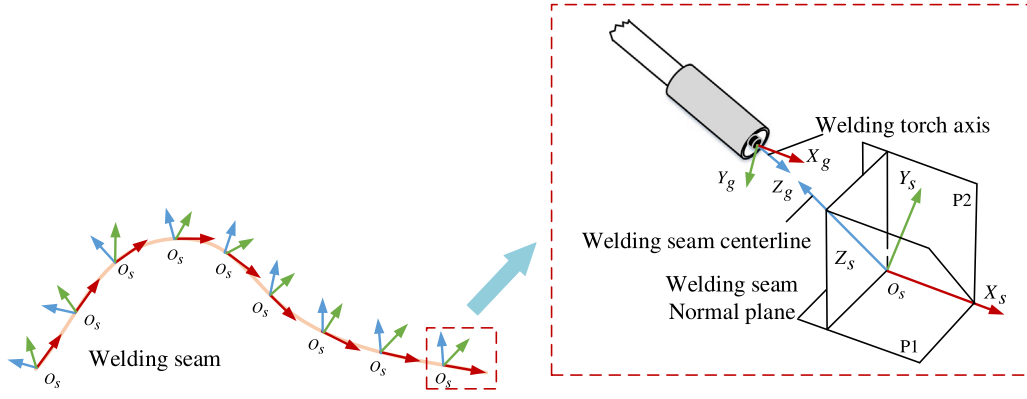


Fig. 24. The dihedral structure.

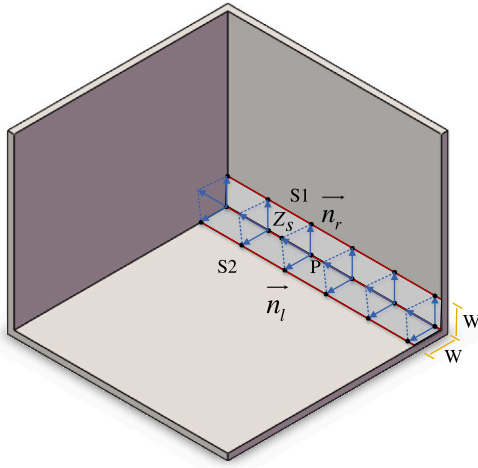


Fig. 25. The solving of welding seam centerline.

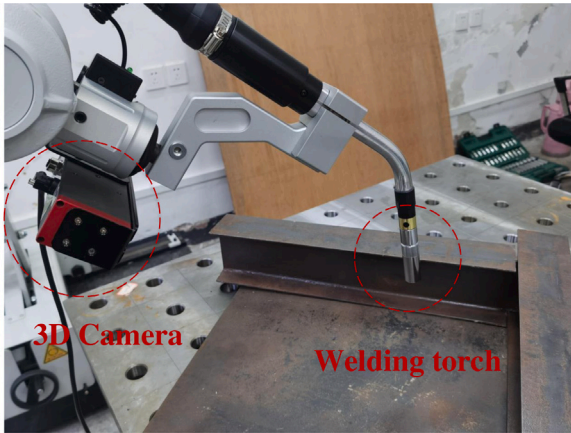


Fig. 26. The experiment platform.

6. Experiments and results

6.1. Method experiment results

Fig. 26 shows the build experiment platform to verify the proposed method. Fig. 27 shows the proposed method experiment results of the three models.

The application results of the proposed method on the three models are relatively good. There is no failure in welding seam extraction

Table 7

The method running time.

Case	Steps	Three-plane butt (ms)	H-section steel butt (ms)	Square tube butt (ms)
1	Form a point cloud model	3025	2384	2520
2	Multiplanes fitting	1247	1826	1935
3	Welding seam extraction	192	223	248
4	Welding path planning	132	127	143
5	Welding posture planning	470	543	623
6	The whole time	5066	5103	5469

Table 8

The experiment error of different models.

Case	Models	Error (mm)	x	y	z
1	Three-plane butt	ME	0.92	0.89	1.19
2	Three-plane butt	RMSE	0.53	0.64	0.79
3	H-section steel butt	ME	1.26	0.77	0.94
4	H-section steel butt	RMSE	0.65	0.49	0.52
5	Square tube butt	ME	0.98	1.22	1.13
6	Square tube butt	RMSE	0.58	0.72	0.68

and welding posture planning. Therefore, the paper method has better robustness.

6.2. Method efficiency analysis

Running time is a pivotal factor to evaluate the method performance. Due to the high requirements for welding efficiency in industrial production, we recorded the average running time of the proposed method for the three models. Based on the laptop with 16 GB memory, the running time results are shown in Table 7.

This method takes about 5 s to plan the welding path of a single medium-thickness plate structural parts model. Therefore, the efficiency of this method can fully adapt to the needs of industrial production.

6.3. Method error analysis

In order to verify the accuracy of the proposed method, we set the welding path of the three models in “teaching-playback” mode as the benchmark to test the precision by calculating the maximum error (ME) and root mean square error (RMSE). Table 8 shows the experiment error results.

As shown in Table 8, the ME between the welding points acquired by the proposed method and manual teaching is less than 1.5 mm, and the RMSE is less than 0.4 mm, which fully meets the welding error requirements. Therefore, the welding path obtained by the proposed method can completely replace the obtained by manual teaching.

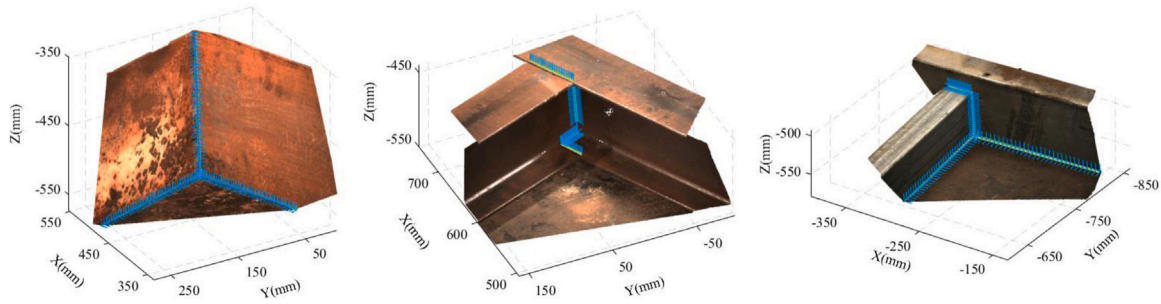


Fig. 27. The paper proposed method experiment results.

Table 9

The efficiency comparison of the two methods.

Case	Methods	Three-plane butt	H-section steel butt	Square tube butt
1	Robotic welding in “teaching-playback” mode	About 8 min	About 9 min	About 9 min
2	The paper method	5066 ms	5103 ms	5469 ms

6.4. Comparison with existing method

Aiming at the problem of robotic welding of medium-thickness plate structural parts, this paper proposes a robotic welding method based on 3D vision. The method can automatically complete the work of welding seam extraction, welding path planning and welding posture planning by processing the point cloud.

To verify the proposed method, we set the “teaching-playback” mode commonly used for robotic welding as the comparison method to illustrate the advantages of the proposed method.

Table 9 shows the comparison results of the two methods. Compared with the robotic welding method commonly used in the industrial field, the efficiency of the proposed method has been significantly improved. Since the “teaching-playback” mode of robotic welding requires manual teaching of the welding path, it consumes a lot of time. In contrast, the proposed method can automatically obtain the welding path through algorithmic processing, which improves the welding efficiency. The method solves the pain point of complex teaching and programming in the robotic welding process. It can automatically complete the welding seam extraction, path planning and posture planning, improving the robotic welding efficiency.

7. Conclusion

From the industrial application perspective, this paper mainly introduces a new welding path planning method based on the 3D vision for robotic and automatic welding of medium-thickness plate structural parts. The method can realize robotic and automatic welding of medium-thickness plate structural parts while eliminating the complicated teaching and programming work in welding path planning. The method has the following advantages.

- (1) The method reduces the dependence on labor in the welding process of medium-thickness plate structural parts, stabilizes and improves the welding quality.
- (2) The method solves the operation difficulties of the traditional programming teaching method and the teaching path deviation caused by the thermal deformation of the welding process.
- (3) The method has a high degree of automation and can simultaneously extract multiple welding seams and perform path and welding posture planning, improving the production efficiency.

CRediT authorship contribution statement

Yusen Geng: Conceptualization, Methodology, Experiment analysis, Writing – original draft, Writing – review & editing. **Min Lai:** Conceptualization, Methodology, Writing – original draft, Experiment analysis. **Xincheng Tian:** Conceptualization, Methodology, Writing – original draft, Supervision. **Xiaolong Xu:** Methodology, Writing – original draft, Writing – review & editing, Supervision. **Yong Jiang:** Data curation, Experiment analysis. **Yuankai Zhang:** Data curation, Experiment analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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